SMART INDIA HACKATHON 2025

- **Problem Statement Title - Wall-Climbing Inspection Robot**
- **Theme - Robotics & Drones**
- **PS Category - Hardware**
- **Team Name - 404 Error Solved**

404 Error Solved

Wall Climbing Inspection Robot

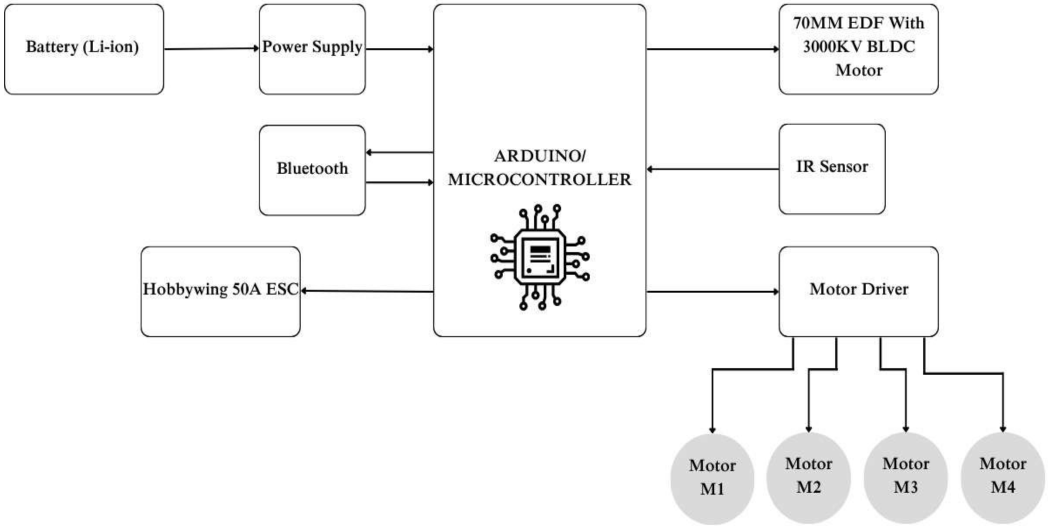
SMART INDIA HACKATHON 2025

PROPOSED SOLUTION	HOW IT ADDRESSES THE PROBLEM	INNOVATION AND UNIQUENESS
• A small robot that sticks to walls and crawls safely. • Records and monitors the surface in real time. • Detects cracks, rust, and leaks during inspection. • Shares a live view with the operator. • Follows a set path for full surface coverage. • Built-in safety features: stops near edges and has a tether to prevent falls	• Keeps humans away from risky heights (no ropes/scaffolding). • Reaches hard-to-access areas quickly (tall buildings, tanks, bridges). • Provides consistent checks with digital records for comparison. • Saves time and cost compared to manual inspection.	• Uses TinyML to detect the onboard crack and will send alert to the user • Unique wall-sticking and crawling mechanism where drones can't work well. • Combines visual inspection along with surface sensing in one pass. • Creates a defect map/report from raw data. • Can adapt to different surfaces (steel, concrete, glass).

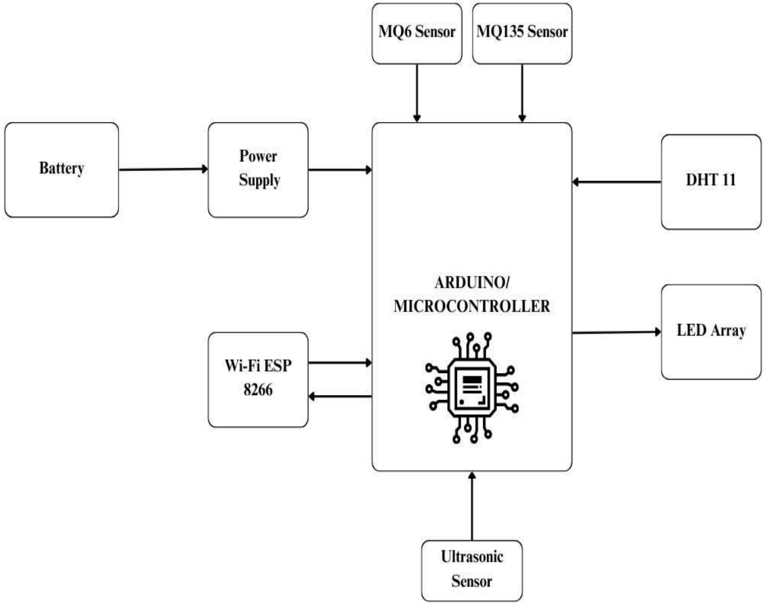
404 Error Solved

TECHNICAL APPROACH

- **Programming Languages**: Arduino IDE, Python (for image processing).
- **Frameworks/Tools**: Arduino IDE, TinyML(for crack detection type).
- **Hardware**: Arduino Nano, ESP32-CAM, L298N Motor Driver, BO DC Motors, BLDC/Vacuum Motor, Li-ion Battery, MQ Sensors, DHT11, Ultrasonic Sensor, Wifi ESP8266, LED Array.



Block Diagram of a Wall Climbing Robot



Block Diagram of Wall Climbing Robot Sensors Integration

404 Error Solved

FEASIBILITY AND VIABILITY

SMART INDIA HACKATHON 2025

Analysis of Feasibility of the idea

- **Technological Viability**: Wall-climbing is achievable using suction/gripper systems with lightweight materials. TinyML enables real-time, on-device crack detection and classification while conserving power.
- **Market Demand**: There is a strong need for safe, automated inspection of bridges, chimneys, dams, and high-rises. Robotic solutions reduce human risk and inspection downtime.
- **Prototype Simplicity**: Open-source hardware/software ensures quick integration and testing. TinyML models can be trained on PCs and deployed easily on ESP32-CAM.
- **Resource Availability**: Components like Arduino, ESP32-CAM, motors, and sensors are low-cost and easily available. Open datasets and free ML libraries make TinyML implementation practical.
- **Scalability & Impact**: The robot can be scaled for city-wide infrastructure monitoring. Over-the-air TinyML updates improve accuracy, ensuring long-term usability.

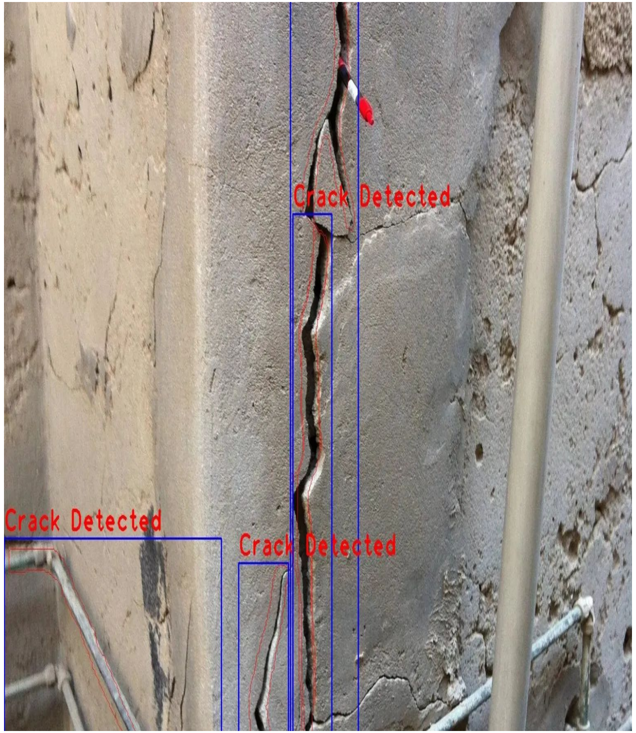
Potential Risks and Strategies Used

- **Adhesion Reliability**: Risk of slipping on uneven walls → Mitigated by hybrid suction and wheel design and lightweight chassis.
- **Power Limitations**: High energy use from motors and ML tasks → Solved with low-power modes, optimized code, and swappable Li-ion batteries.
- **Model Accuracy**: TinyML may misclassify cracks → Improved by training with diverse datasets and OTA model updates.
- **Environmental Factors**: Dust, poor lighting, or weather interference → Overcome with LED/IR illumination, protective casing, and sensor fusion.
- **Mechanical Durability**: Wear on motors/suction units → Addressed using robust components, modular design, and periodic calibration.

404 Error Solved

IMPACT AND BENEFITS

- **Enhanced Safety**: Removes human risk by deploying robots in hazardous inspection zones.
- **Improved Efficiency**: Conducts faster, frequent, and automated inspections, minimizing downtime.
- **Cost-Effective**: Cuts expenses on scaffolding, cranes, and manual labor.
- **High-Quality Insights**: Delivers real-time video, crack classification via TinyML, and environmental sensing for accurate decisions.
- **Versatility**: Adaptable for infrastructure monitoring, solar panel/window cleaning, defense, and disaster response.
- **Sustainability**: Enables early fault detection, lowering repair costs and preventing material wastage.



404 Error Solved

RESEARCH AND REFERENCES

SMART INDIA HACKATHON 2025

- https://hausbots.com/wall-climbing-robots/
- https://www.sciencedirect.com/science/article/pii/S2773186324000781
- https://www.mdpi.com/2075-1702/13/6/521
- (PDF) Design and Analysis of Wall-Climbing Robot
- Design and Implementation of ESP32-Based Edge Computing for Object Detection
- 2016 - MED - A Survey on Pneumatic Wall-Climbing Robots for Inspection
- Wall Climbing Robots - HausBots
- [2301.05882] Design and Development of Wall Climbing Robot
- https://www.instructables.com/GECO-Wall-Climbing-Robot/